

Newton's laws of motion:

First law: An object will continue to be at rest or in a state of uniform motion in a straight line unless acted upon by an external force.

↳ This introduces the concept of inertial & non-inertial frame of reference.

Second law: When a $\vec{F}$ is applied on mass m , the rate of change of its velocity is proportional to the applied force.

$$\vec{F} \propto m \frac{d\vec{v}}{dt}$$

$$\vec{F} = m \vec{a}$$

→ It can be written as $\vec{F} = \frac{d\vec{p}}{dt}$ where $\vec{p} = m\vec{v}$ is the momentum of the particle.

Third law: When a body of mass m_1 , exerts a force on another body of mass m_2 . The body m_1 , experiences a force equal in magnitude & opposite in direction.

→ Action is on body 2; reaction is on body 1 & the line of action is the same for both these forces.

→ constant acceleration:

②

1) $\vec{a} = \text{constant}$

$$\Rightarrow \frac{d\vec{v}}{dt} = \vec{a}$$

$$\int_u^v d\vec{v} = \int_0^t \vec{a} dt$$

$$\vec{v} - \vec{u} = \vec{a}t$$

$$\underline{\underline{\vec{v} = \vec{u} + \vec{a}t}}$$

2) $\vec{v} = \vec{u} + \vec{a}t$

$$\frac{dx}{dt} = u + at$$

$$\int_0^s dx = \int_0^t u dt + \int_0^t at dt$$

$$\underline{\underline{s = ut + \frac{1}{2}at^2}}$$

3) $\frac{d\vec{v}}{dt} = \vec{a}$

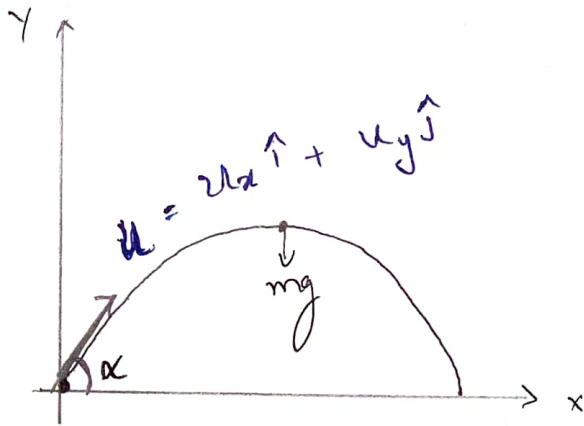
$$\frac{d\vec{v}}{dx} \cdot \frac{dx}{dt} = \vec{a}$$

$$\vec{v} d\vec{v} = a dx$$
$$\int_u^v \vec{v} d\vec{v} = \int_0^s a dx$$

$$\frac{v^2}{2} - \frac{u^2}{2} = as \Rightarrow \underline{\underline{v^2 - u^2 = 2as}}$$

→ Projectile motion:

3



→ neglect air resistance.

$$\vec{F} = F_x \hat{i} + F_y \hat{j}$$

$$\vec{F} = 0 + (-mg) \hat{j} = \underline{\underline{-mg \hat{j}}}$$

$$\vec{u} = \text{initial velocity} = u_x \hat{i} + u_y \hat{j}$$

$$\underline{\underline{\vec{u} = u \cos \alpha \hat{i} + u \sin \alpha \hat{j}}}$$

$$m \vec{a} = -mg \hat{j}$$

$$m \frac{dv_y}{dt} = -mg \hat{j}$$

$$\frac{dv_y}{dt} = -g \rightarrow \int_{u_y}^{v_y} dv_y = \int_0^t -g dt$$

$$v_y = u_y - gt$$

$$\underline{\underline{v_y = u \sin \alpha - gt}}$$

$$v_x = u \cos \alpha \quad \text{constant}$$

$$\therefore x = u \cos \alpha t \quad \left. \begin{array}{l} \text{constant velocity} \\ \text{case} \end{array} \right\}$$

$$\underline{\underline{x = u \cos \alpha t}}$$

$$v_y = u \sin \alpha - gt$$

$$\frac{dy}{dt} = u \sin \alpha - gt$$

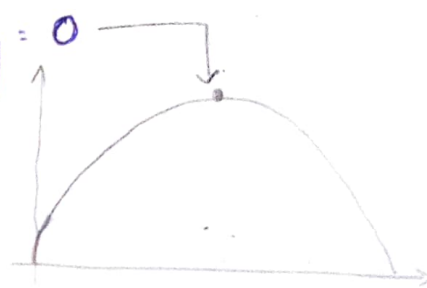
$$\underline{\underline{y = u \sin \alpha t - \frac{1}{2}gt^2}}$$

At maximum ~~value~~ height $v_y = 0$

$$\Rightarrow v_y = u \sin \alpha - gt$$

$$\therefore 0 = u \sin \alpha - gt$$

$$t = \frac{u \sin \alpha}{g}$$



Time to reach maximum ht.

$$\therefore \text{total time of flight} = 2t$$

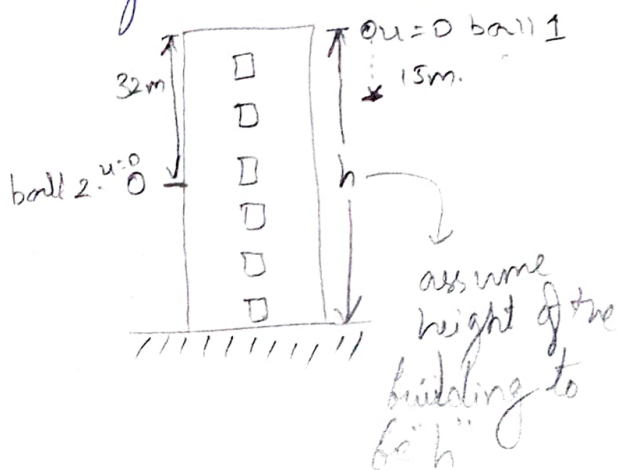
$$\underline{\underline{t_{\text{total}} = \frac{2u \sin \alpha}{g}}}$$

$$\begin{aligned}
 \text{Range} &= x_{\text{total}} = u \cos \alpha \cdot t_{\text{total}} \\
 &= u \cos \alpha \cdot \left(\frac{2 \sin \alpha \cdot u}{g} \right) \\
 &= \frac{u^2 2 \sin \alpha \cos \alpha}{g}
 \end{aligned}$$

$$\boxed{\text{Range} = x_{\text{total}} = \frac{u^2 \sin 2\alpha}{g}}$$

- 1) Two people are playing over a multi-storey building. One is at the topmost floor and the other is at a floor 32 m below the top. When a ball is dropped from the top of the building by first person, and this ball reaches 15 m from top, the second person drops a ball. If both the balls hit the ground floor at the same time, determine:
- height of the building
 - velocity of both balls as they hit the ground
 - Time for both balls to reach the ground.

→ Neglect air resistance.



For both balls, the initial velocity is "0" $\therefore$ it says they were dropped. Take $g = 9.81 \text{ m/s}^2$ (Constant accn. case).

6
Sol. For the first ball:

$$u = 0 \quad h_1 = 15 \text{ m.}$$

using $v^2 - u^2 = 2gh_1$

we get $v^2 - 0^2 = 2g \cdot 15$

$$v = \sqrt{2 \times 9.81 \times 15} = \underline{\underline{17.16 \text{ m/s}}}$$

This instant is when the second ball is dropped
for second ball:

using $s = ut + \frac{1}{2}gt^2$ we can write as

$$h - 32 = 0 + \frac{1}{2}gt^2$$

For first ball a similar equation can be written

$$h - 15 = ut + \frac{1}{2}gt^2$$

$$h - 15 = 17.16t + \frac{1}{2} \cdot 9.81 t^2$$

$$- h - 32 = \frac{1}{2} \cdot 9.81 t^2$$

} Subtract
these 2 eqns.

$$32 - 15 = 17.16t$$

$$t = \frac{17}{17.16} = \underline{\underline{0.99 \text{ s}}}$$

$$\therefore h - 32 = \frac{1}{2} g t^2$$

$$h - 32 = \frac{1}{2} \times 9.81 \times (0.99)^2$$

$$h = 32 + 4.81$$

$$\underline{h = 36.81 \text{ m}}$$

$\therefore$ The height of the building is 36.81 m

→ For the ball 1

$$v^2 = u^2 + 2gh$$

$$v^2 = 0 + 2 \times 9.81 \times 36.81$$

$$v = \sqrt{2 \times 9.81 \times 36.81} = 26.87 \text{ m/s}$$

velocity of ball 1 before hitting the floor is 26.87 m/s

→ for the ball 2

$$v^2 = u^2 + 2g(h - 32)$$

$$v^2 = 0 + 2 \times 9.81 \times (h - 32)$$

$$v^2 = 2 \times 9.81 \times (36.81 - 32)$$

$$v = \sqrt{2 \times 9.81 \times 4.81} = 9.71 \text{ m/s}$$

velocity of ball 2 before hitting the floor is

26.87 m/s

→ Time taken by ball 1 to hit the floor is (8)

$$v = u + gt_1$$

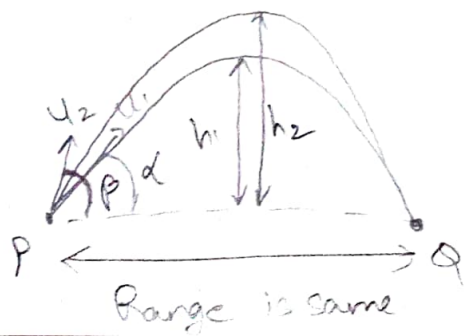
$$26.87 = 0 + 9.81 \cdot t_1$$

$$t_1 = \underline{2.74 \text{ s}}$$

→ Time taken by ball 2 to hit the floor is already calculated as $t_2 = \underline{0.99 \text{ s}}$

- 2) A fielder P throws a ball to another fielder Q. If the ball is thrown 2 times with same velocity and for the same distance between P and Q. Neglecting air resistance, determine:
- The relationship between their distance & the maximum heights attained by the ball in the 2 scenarios.
 - The relationship between angle of projection & the maximum heights.

sol Let α be the angle of the first projectile motion & β be the angle of the second projectile motion.



Given $u_1 = u_2$

Range "PQ" is constant

Let h_1 & h_2 be maximum heights for ~~both~~ ^{the} projectiles.

∴ Projectile range is same.

$$R = \frac{u^2 \sin 2\alpha}{g} \quad \left\{ \text{for first projectile motion.} \right.$$

$$= \frac{u^2 \sin 2\beta}{g} \quad \left\{ \text{for second projectile motion.} \right.$$

$$\therefore \frac{u^2 \sin 2\alpha}{g} = \frac{u^2 \sin 2\beta}{g}$$

$$\Rightarrow \sin 2\alpha = \sin 2\beta \quad \because \sin 2\alpha = \sin 2(90 - \alpha)$$

$$\Rightarrow \underline{\underline{\beta = 90 - \alpha}} \quad \leftarrow$$

For the projectiles to have same range the 2 angles of projection should be α & $90 - \alpha$.

Now for projectile 1 the height h_1 can be written as

$$h_1 = \frac{u^2 \sin^2 \alpha}{2g}$$

For projectile 2 the height h_2 will be

$$h_2 = \frac{u^2 \sin^2 \beta}{2g} = \frac{u^2 \sin^2 (90 - \alpha)}{2g} = \frac{u^2 \cos^2 \alpha}{2g}$$

Now multiply $\{ h_1 \times h_2$

$$\begin{aligned}
 \therefore h_1 \times h_2 &= \frac{u^2 \sin^2 \alpha}{2g} \cdot \frac{u^2 \cos^2 \alpha}{2g} \\
 &= \left(\frac{u^2}{2g} \right)^2 (\sin \alpha \cos \alpha)^2 \\
 &= \left\{ \left(\frac{u^2}{2g} \right) \left(\frac{\sin 2\alpha}{2} \right) \right\}^2 \\
 &= \left\{ \frac{1}{4} \cdot \left(\frac{u^2 \sin 2\alpha}{g} \right) \right\}^2
 \end{aligned}$$

$\rightarrow$ This is the Range PQ

$$h_1 h_2 = \left\{ \frac{1}{4} \cdot R \right\}^2$$

$$\therefore R = 4 \sqrt{h_1 h_2}$$

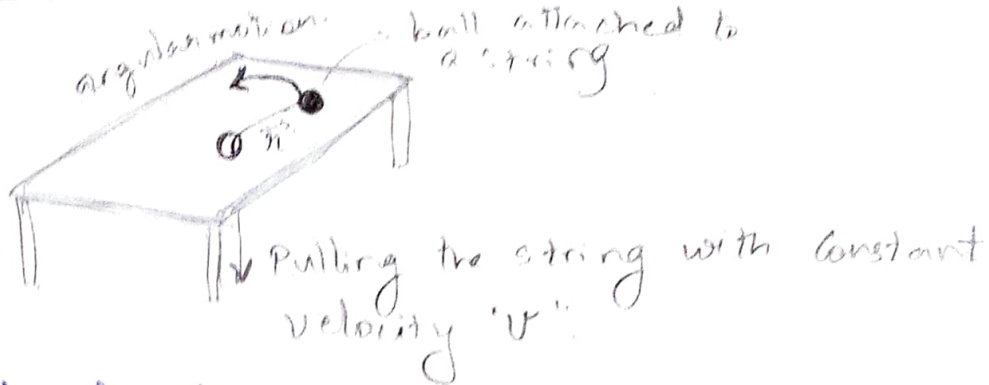
$\rightarrow$ Relationship between Range R , h_1 & h_2 of the two projectiles.

$$\text{Now divide } \left\{ \frac{h_1}{h_2} = \frac{\frac{u^2 \sin^2 \alpha}{2g}}{\frac{u^2 \cos^2 \alpha}{2g}} = \left(\frac{\sin \alpha}{\cos \alpha} \right)^2 \right.$$

$$\therefore \tan \alpha = \sqrt{\frac{h_1}{h_2}} \Rightarrow \alpha = \tan^{-1} \left(\sqrt{\frac{h_1}{h_2}} \right)$$

Relationship between heights $\underline{h_1, h_2}$ & $\underline{\alpha}$

3)



(11)

Given a frictionless table on which a ball is moving as shown in the figure above. Assume no external force is acting on the ball in the $\hat{\phi}$ direction. A string attached to ball is being pulled through a hole on the table in the $\hat{r}$ direction with a constant velocity " v ". Find the force need to pull the string with this constant velocity " v ". Let " m " be the mass of the ball. Given initial conditions are at $t=0$ $r=r_0$ & $\omega=\omega_0$.

Ans.Sol.

The acceleration of particle in polar coordinates are :

$$\vec{a} = (a_r - r\omega^2)\hat{r} + (2v_r\omega + r\alpha)\hat{\phi}$$

$$\text{where } a_r = \frac{dv_r}{dt} \quad v_r = \frac{dr}{dt} \quad \omega = \frac{d\phi}{dt} \quad \alpha = \frac{d\omega}{dt}$$

$\therefore$ the table is frictionless & it is given that no external force is in $\hat{\phi}$ direction we can say that the acceleration in $\hat{\phi}$ direction is zero.

⇒ only radial component of acceleration remains

$$\therefore \vec{a} = (a_r - r\omega^2) \hat{r}$$

Now Given that "V" is the constant velocity in radial direction $\Rightarrow v_r = \dot{r} = \text{const}$

$$\therefore a_r = \frac{dv_r}{dt} = \frac{dV}{dt} = 0 \rightarrow \because V \text{ is constant}$$

$\therefore$ Net acceleration on the ball is

$$\underline{\underline{\vec{a} = -r\omega^2 \hat{r}}}$$

$\therefore$ Force required to pull the string at const. velocity V is $\vec{F} = m\vec{a}$

$$\Rightarrow \vec{F} = m \cdot (-r\omega^2) \hat{r}$$

$$\boxed{\vec{F} = -m r \omega^2 \hat{r}}$$

→ To further analyze this using initial conditions we use the fact that the acceleration in $\hat{\phi}$ direction is zero.

⇒

$$2 v_r \omega + r \alpha = 0$$

multiply with r on both sides

$$2 r v_r \omega + r^2 \alpha = 0$$

$$2 r_1 v_{r_1} \omega + r_1^2 \alpha = 0$$

same expression.

$$\frac{d}{dt} (r^2 \omega) = 0$$

Expanding this gives

$$\frac{d(r^2)}{dt} \omega + r^2 \frac{d\omega}{dt}$$

$$2 r_1 \frac{dr_1}{dt} \omega + r_1^2 \alpha$$

$$\underline{2 r_1 v_{r_1} \omega + r_1^2 \alpha}$$

$$\therefore \frac{d}{dt} (r^2 \omega) = 0$$

$$\Rightarrow r^2 \omega = \text{Constant.}$$

$$\therefore t=0 \quad r = r_0 \text{ \& } \omega = \omega_0$$

$$\text{at } t=0 \quad r_0^2 \omega_0 = \text{Constant}$$

$$\text{at } t=t \quad r^2 \omega = \text{Constant}$$

$$\Rightarrow r^2 \omega = r_0^2 \omega_0$$

but $r = r_0 - vt$ } the radius of the string changes with a constant velocity.

$$\therefore (r_0 - vt)^2 \omega = r_0^2 \omega_0$$

$$\omega = \frac{r_0^2 \omega_0}{(r_0 - vt)^2}$$

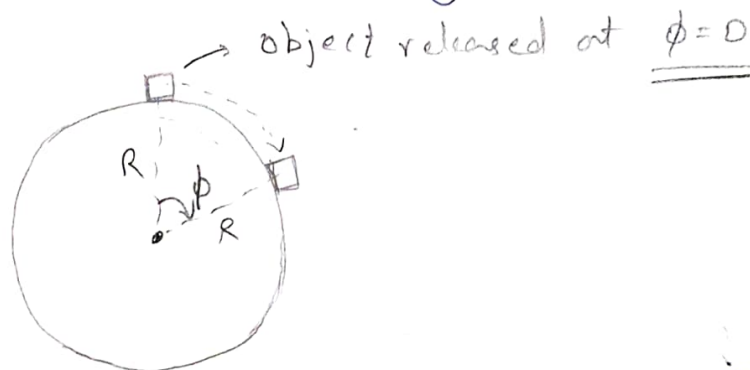
use this in $\vec{F} = -m r \omega^2$

$$\vec{F} = -m (r_0 - vt) \left(\frac{r_0^2 \omega_0}{(r_0 - vt)^2} \right)^2$$

$$\therefore \vec{F} = \frac{-mg_0^4 \omega_0}{(g_0 - vt)^3}$$

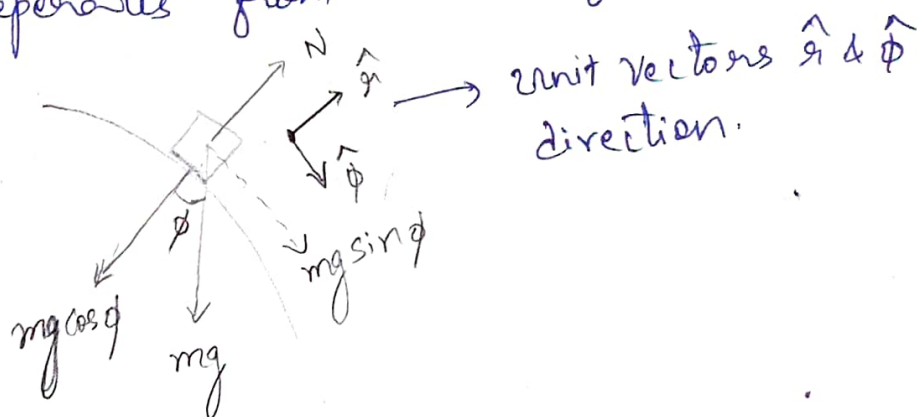
This is the force required to pull the wire with a constant velocity " v ".

4)



A small object is at rest on top of a frictionless cylindrical surface. The object is given a slight nudge to the right so that it slides along the surface. Determine the angle ϕ at which the object separates from the surface.

Sol



forces in radial $\left. \begin{array}{l} \text{ } \\ (\hat{r}) \text{ direction} \end{array} \right\} \bullet N - mg \cos \phi = m a_r$

forces in angular $\left. \begin{array}{l} \text{ } \\ (\hat{\phi}) \text{ direction} \end{array} \right\} mg \sin \phi = m a_\phi \text{ (no friction)}$

The acceleration in polar coordinates is given as

$$\vec{a} = (a_r - r\omega^2) \hat{r} + (2v_r\omega + r\alpha) \hat{\phi}$$

$$\therefore r = R = \text{constant}$$

$$v_r = \frac{dr}{dt} = 0$$

$$\therefore a_r = \frac{dv_r}{dt} = 0$$

$$\Rightarrow \vec{a} = -r\omega^2 \hat{r} + r\alpha \hat{\phi}$$

$$\Rightarrow a_r = -r\omega^2 \quad a_\phi = r\alpha$$

$$\therefore N - mg \cos \phi = ma_r = -mr\omega^2 \quad \left. \vphantom{\begin{matrix} N - mg \cos \phi \\ = ma_r \end{matrix}} \right\} \text{ forces in } \hat{r} \text{ direction.}$$

$$mg \sin \phi = ma_\phi = mr\alpha \quad \left. \vphantom{mg \sin \phi} \right\} \text{ forces in } \hat{\phi} \text{ direction.}$$

$$\Rightarrow g \sin \phi = r\alpha$$

$$\alpha = \frac{g \sin \phi}{r}$$

$$\text{but } \alpha = \frac{d\omega}{dt} \Rightarrow \frac{d\omega}{dt} = \frac{g \sin \phi}{r}$$

$$\frac{d\omega}{d\phi} \cdot \frac{d\phi}{dt} = \frac{g \sin \phi}{r} \quad \left. \vphantom{\frac{d\omega}{d\phi} \cdot \frac{d\phi}{dt}} \right\} \text{ using chain rule.}$$

ω

$$\therefore \int \omega \cdot d\omega = \int \frac{g \sin \phi}{r} d\phi$$

Given that at $\phi = 0$ $\omega = 0$ ($\because$ object released at $\phi = 0 \Rightarrow$ initial ~~angular~~ velocity is 0 $\Rightarrow \omega = 0$) $\therefore$

$$\therefore \int_0^{\omega} \omega d\omega = \frac{g}{r} \int_0^{\phi} \sin \phi d\phi$$

$$\frac{\omega^2}{2} = -\frac{g}{r} \cos \phi \Big|_0^{\phi} = -\frac{g}{r} (1 - \cos \phi)$$

$$\therefore \omega^2 = \frac{2g}{r} (1 - \cos \phi)$$

using

$$N - mg \cos \phi = -m r \omega^2$$

$$N = mg \cos \phi - m r \omega^2$$

$$N = mg \cos \phi - m r \cdot \frac{2g}{r} (1 - \cos \phi)$$

$$N = mg \cos \phi - 2mg + 2mg \cos \phi$$

$$N = 3mg \cos \phi - 2mg$$

for the object to leave the surface $N = 0$

$$\therefore 3mg \cos \phi = 2mg \Rightarrow \cos \phi = \frac{2}{3} \therefore \phi = \cos^{-1}\left(\frac{2}{3}\right)$$